

AGENTIC AI – RESUME SCREENER AND CANDIDATE MATCHER

PAGE 1 – INTRODUCTION

1. Introduction

Recruitment is an important process in every organization. Companies may receive a large number of resumes for a single job position. Manually reading every resume and comparing it with job requirements can be time-consuming and repetitive for recruiters.

The proposed **Agentic AI – Resume Screener and Candidate Matcher** is an intelligent recruitment-support application that uses Artificial Intelligence to assist with the initial screening of candidates. The system takes a candidate resume and a Job Description (JD), extracts useful information, compares the candidate with the job requirements, and produces an understandable matching report.

2. Problem Statement

Traditional resume screening mainly depends on manual review. When hundreds of resumes are received, recruiters may spend significant time identifying relevant skills, education, projects, and experience. It can also be difficult to consistently compare every candidate with the same job requirements.

Therefore, an AI-based system is proposed to automate the initial screening process and provide structured information that can help recruiters review candidates more efficiently.

3. Objectives

- Extract important information from candidate resumes.
- Analyze Job Descriptions and identify important requirements.
- Compare candidate skills and qualifications with job requirements.
- Identify matched and missing skills.
- Generate a candidate-job matching score.
- Provide an understandable explanation of the matching result.
- Support recruiters during the initial candidate review process.

4. Target Users

- HR professionals
- Recruiters and hiring teams
- Placement officers
- Small and medium-sized organizations
- Job seekers for self-evaluation

PAGE 2 – PROJECT OVERVIEW

1. Overview

The Resume Screener and Candidate Matcher is designed as a multi-agent AI workflow. Instead of using one AI component for the complete task, the system divides the work into smaller specialized tasks. Each agent performs a particular function and passes its result to the next stage.

2. Agentic AI Concept

Agentic AI refers to an AI system that can perform a sequence of related tasks toward a goal. In this project, specialized agents cooperate to complete the resume screening and candidate matching workflow.

3. Main Agents

Resume Analysis Agent: Reads the extracted resume text and identifies candidate name, education, skills, experience, projects, and certifications.

Job Description Agent: Reads the job description and identifies the job role, required skills, preferred skills, education, and experience requirements.

Candidate Matching Agent: Compares candidate information with the job requirements and identifies matched and missing skills.

Evaluation Agent: Analyzes the matching information and produces a structured evaluation using predefined criteria such as skill, education, experience, and project relevance.

Recommendation Agent: Generates a clear summary for further human review based on the matching evidence.

4. System Workflow

Step	Process
1	Upload Resume + Enter Job Description
2	Extract Resume Text
3	Resume Agent analyzes candidate
4	Job Agent analyzes requirements
5	Matching Agent compares both
6	Evaluation Agent analyzes match
7	Recommendation Agent generates report

5. Expected Result

For example, if a job requires Python, SQL, Flask, REST API and Git, the system can identify which of these skills are present in a candidate's resume, which are missing, and provide a matching score with an explanation.

Note: The system is intended as a decision-support tool. The final hiring or selection decision should remain with a human recruiter.

PAGE 3 – PRE-PHASE

1. Pre-Phase

The pre-phase covers the activities performed before implementation. It includes understanding the problem, identifying users, defining inputs and outputs, deciding the workflow, and selecting suitable technologies.

2. Problem Identification

- Recruiters may receive a large number of resumes.
- Manual screening requires time and effort.
- Important candidate skills can be difficult to identify quickly.
- Comparing every resume with a Job Description is repetitive.
- Missing skills may not be immediately visible.
- Recruiters need structured and explainable candidate information.

3. Proposed Solution

The proposed system uses Agentic AI to divide the screening process into multiple stages. Resume information is extracted first, job requirements are analyzed separately, and then both sets of information are compared. The final result contains matched skills, missing skills, candidate details, and an explanation.

4. Inputs

- Candidate resume in PDF or DOCX format.
- Job Description provided by the recruiter.
- Optional job-specific criteria such as experience or preferred skills.

5. Outputs

- Candidate name and profile information.
- Education, skills, experience, projects, and certifications.
- Matched skills.
- Missing skills.
- Candidate-job match score.
- AI-generated explanation.
- Recommendation for further human review.

6. Scope

The initial version focuses on resume processing, Job Description analysis, candidate matching, and report generation. Future versions can include multiple resume comparison, semantic matching, dashboards, interview question generation, and resume improvement suggestions.

7. Feasibility

The project can be developed using commonly available software tools and cloud-based AI APIs. A normal laptop with Python, VS Code, an internet connection, and the required libraries is sufficient for development.

PAGE 4 – REQUIREMENTS

1. Functional Requirements

- **Resume Upload:** The system should accept supported resume files such as PDF and DOCX.
- **Text Extraction:** The system should extract readable text from uploaded resumes.
- **Resume Analysis:** The system should identify name, education, skills, experience, projects, and certifications.
- **Job Analysis:** The system should identify job role, required skills, preferred skills, education, and experience.
- **Candidate Matching:** The system should compare candidate information with job requirements.
- **Missing Skill Detection:** The system should identify important job skills not found in the candidate profile.
- **Match Score:** The system should generate a score using defined matching criteria.
- **Explainable Report:** The system should provide reasons and evidence for the matching result.
- **Multiple Candidates:** The system should be extendable to screen multiple resumes for one job.

2. Non-Functional Requirements

- **Usability:** The interface should be simple and easy for recruiters to use.
- **Performance:** Resume analysis should complete within a reasonable time.
- **Reliability:** The application should handle invalid files and API errors.
- **Security:** Candidate personal information and uploaded resumes should be handled securely.
- **Maintainability:** Agents and services should be separated into modules.
- **Explainability:** Results should be understandable rather than showing only a score.

3. Technology Requirements

Component	Technology
Language	Python
Backend	Flask
AI / LLM	Groq API
PDF Processing	PyMuPDF
DOCX Processing	python-docx
Matching	Python + Embeddings
Database	SQLite
Frontend	HTML, CSS, JavaScript
Configuration	.env
Development Tool	VS Code

4. Hardware Requirements

- Laptop or desktop computer.
- Minimum 4 GB RAM; 8 GB recommended.
- At least 2 GB free storage.
- Internet connection for cloud-based AI API access.

5. Software Requirements

Python 3.x, VS Code, pip, Flask, Groq SDK, PyMuPDF, python-docx, python-dotenv, and SQLite.

6. Conclusion

The Agentic AI Resume Screener and Candidate Matcher provides a structured approach to initial candidate screening. By combining specialized AI agents with document processing and matching techniques, the system can reduce repetitive screening work and provide useful, explainable information to recruiters while keeping final employment decisions under human control.